

CSE 412 final project

Universities and Features analysis in USA

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Introduction:

A university is a higher education institution that offers degree in various fields of study.

Universities can be public or private and were founded in around 17th centuries.

Therefore, educational organizations gather extensive and detailed information about universities based on various attributes.

Many students face challenges in selecting universities due to the abundance of reputable institutions in the U.S. Often, they lack information about each university's location and application competitiveness. Our visualizations aim to bridge this gap by offering a clear and concise visual representation of essential data, enabling students and parents to better understand their options and make more informed choices.

In our report, we will discuss at least three questions: 1: how are the universities distributed in states and what is the average tuition. 2: What is the relationship among ranks, SAT scores, and graduation requirements. 3: What is the relationship among tuition, acceptance rate, and education resources.

IMPORT DATA:

```
universityData = ▶ Array(278) [Object, Object, Object, Object, Object, Object, Object, Objec
```

Dataset Description:

Overview:

The university dataset contains **277 universities** in USA with key attributes such as rankings, names of universities, tuition fees, SAT scores, acceptance rates, and student-to-faculty ratios. This dataset are collected from College Evaluator. This data is real-time updated. The dataset we used to analyze was collected in Feb 17th.

https://www.collegeevaluator.com/rankings/national-universities/#google_vignette

Columns and Data Types:

Rank – The ranking of the university.

College Name – The name and location of the university.

Undergraduate Tuition & Fees (\$) – The annual tuition and fees for undergraduate

programs.

Graduate Tuition & Fees (\$) – The annual tuition and fees for graduate programs.

SAT Score – The SAT scores of admitted students.

Acceptance Rate – The proportion of applicants admitted.

Population – The total number of students enrolled.

Graduation Rate – The percentage of students who graduate.

Student-to-Faculty Ratio – The ratio of students to faculties.

Data Integrity Observations:

The dataset contains 279 rows and 9 columns, where the first row is the head of table, and the last row is the average values of numerical attributes. The "SAT Score" and "Acceptance Rate" columns, has some missing values, which might because the school do not report the related information to public.

Data formatting issues: "College Name" includes both the university name and its location. "SAT Score" and "Acceptance Rate" contain '%' which takes problem to analysis it as numerical values, so we convert it to decimal number without '%' sign, and marked it in the name of the attributes. "Undergraduate Tuition & Fees (\$)" and "Graduate Tuition & Fees (\$)" contain dollar '\$' sign in the original dataset, so we convert it as numerical values excluding dollar signs.

dataset imported specification:

1:national_universities_rankings@2.csv

This dataset includes all information from the data resources and corrects the faculty to student ratio from X to Y to percentage format, facilitating the table generation.

2:state_centroid.csv

This dataset has the state name and their centroid coordinate, which helps the question1 map plot.

3:university@1.csv

This dataset includes all information from the data resources and one more state column, showing which state those universities located.

Analysis

Visualization 1

```
uni = ▶ Array(277) [Object, Object, Object, Object, Object, Object, Object, Object, (
```



A bubble map visualizing the distribution of universities across the United States based on

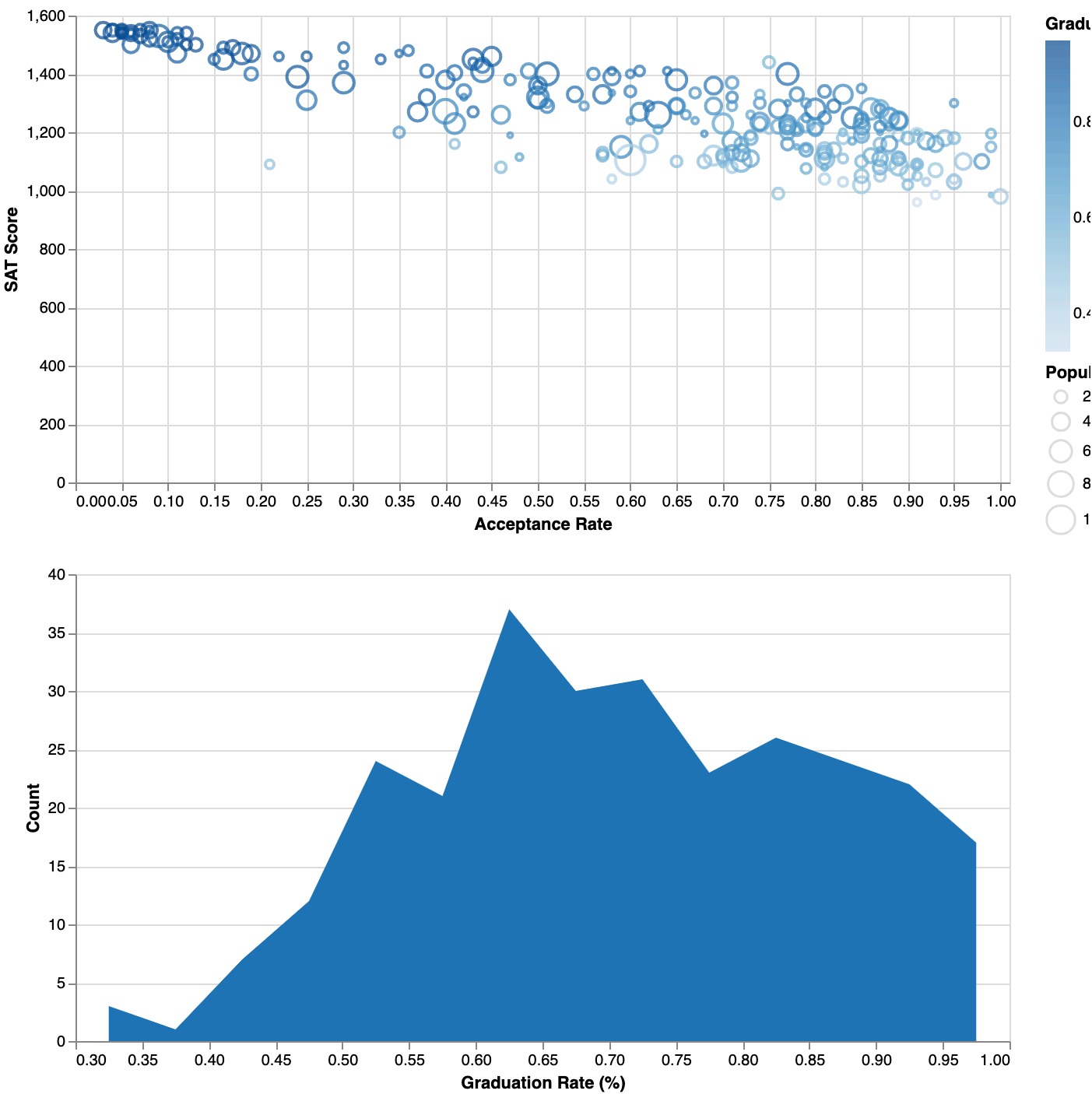
tuition costs at higher education institutions, with different states represented by bubbles of varying sizes and shades of blue.

Bubble Color: Represents the average undergraduate tuition fees in each state, with darker blue indicating higher tuition. Bubble Size: Reflects the number of schools in each state, with larger bubbles representing a higher number of institutions. Legend: Tuition Fee Scale (Color Gradient): Ranges from \$20,000 (lighter blue) to \$60,000 (darker blue). School Count (Bubble Size): Categorized as 5, 10, 15, or 20 schools in a state.

States in the Northeast and California have the highest tuition fees, indicated by darker blue bubbles. States with a higher number of universities include New York, Massachusetts, Pennsylvania, and California, as reflected in the larger bubbles. The Midwest and Southern states tend to have lower average tuition fees, with lighter blue shades. The West Coast, particularly California, has both high tuition fees and a significant number of institutions. The distribution suggests that states with a higher concentration of prestigious universities tend to have higher average tuition fees.

Visualization 2

University Data with Graduation Rate Distribution



Description:

An interactive visualization exploring the relationships between university acceptance rates, SAT scores, and graduation rates, with a dynamic filtering feature that allows users to adjust the selection region in the scatter plot, updating the graduation rate distribution below accordingly.

Features:

The Plot (Scatter Plot): X-axis: University Acceptance Rate (ranging from 0% to 100%) Y-axis: SAT Score (ranging from 0 to 1600) Color: Graduation Rate (ranging from 0.4 to 0.8) Size: Population (ranging from 2 to 10)

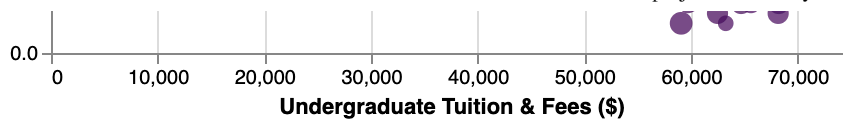
Top Plot (Scatter Plot): X-axis: University acceptance rate (ranging from 0% to 100%). Y-axis: SAT scores of admitted students (ranging from 200 to 1600). Bubble Size: Represents total student population at the university (larger bubbles indicate higher enrollment). Bubble Color: Encodes graduation rate, with darker shades representing higher rates. Interactive Selection: Users can select a region in the scatter plot, dynamically updating the histogram below. Bottom Plot (Histogram): X-axis: Graduation rate (%) of universities. Y-axis: Count of universities in each graduation rate bin. Dynamic Updates: The histogram reflects only the universities selected in the scatter plot above, adjusting based on user selection.

Insights:

Universities with lower acceptance rates (left side of the scatter plot) tend to have higher SAT scores and higher graduation rates (darker bubbles). Institutions with higher acceptance rates (right side) show a wider spread of SAT scores and graduation rates, with some lower-performing schools having lighter-colored bubbles. The graduation rate distribution below shifts based on selection, allowing users to focus on different university tiers and observe how graduation rates vary.

Visualization 3:





Tuition fee in Dollars(\$0 - \$75,000):

Acceptance Rate(0.0 - 1.0) above:

Description:

An interactive bubble chart that allows users to explore relationships between tuition fees, acceptance rates, student-faculty ratios, and university rankings, showing detailed information in the floated block.

Features:

X-axis: Undergraduate tuition fees (ranging from \$0 to \$75,000).

Y-axis: Acceptance rate (0% to 100%).

Bubble Size: Represents the student-to-faculty ratio (larger bubbles indicate higher faculty-to-student ratios).

Bubble Color: Encodes ranking groups, where universities are grouped into bins of 50 ranks.

Interactive Filters:

Tuition Fee Slider: Users can adjust the maximum of undergraduate tuition fees.(\$75,000 in default)

Acceptance Rate Slider: Users can filter universities based on minimum acceptance rates. (0% in default)

Insights:

In general, small bubbles in purple cluster in the lower right corner, and relatively large bubbles appear to be on the top left side, which means that universities in higher rankings tend to be more expensive, lower acceptance rate, lower student-to-faculty ratio, and higher SAT score from detailed information, vice versa. The insights conform to the common sense that higher ranking universities have better education resources that students in those universities tend to receive lots of individualized attention and hands on learning, and at the same time tuition at top universities in the United States are expensive (over \$60,000).

Importing cells

```
import {p5} from "@tmcw/p5"
```

```
import {printTable} from "@uwdata/data-utilities"
```

```
import {render} from "@vega/vega-lite-api-v5"
```

```
import {vl} from "@vega/vega-lite-api-v5"
```

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```
import {vl} from "@vega/vega-lite-api-v5"
```

Link:

<https://observablehq.com/d/ab87686200da7589>

Reflection:

In FP2, we've learned deeply on how to make interactive visualizations in Observable using Vega Lite. We learned how to implement json files to make geographic map visualizations, and Coordinated brushing for Interactive filtering, making it a more interactive and insightful data exploration tool. Each visualization could answer one or more questions in detail at the same time leading the importance of the interactive visualization. One of the challenges we have was how to deal with the dataset and convert it from string with special signs to desired values. Another challenge I encountered was applying numerical values (average tuition fee per state) onto the map to effectively show tuition fee differences across the U.S. This process required multiple steps, including data preparation, geographical mapping, and encoding numerical values visually. Challenge in designing effective visualizations was deciding which attributes to display. While it's tempting to include as much information as possible, I learned that too many variables can confuse viewers and make insights harder to interpret.